

SQL exercise 1

Employees table

id	first-name	last-name	department	salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Retrieve all columns from the employees table

* Select *
FROM employees;

id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
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3	Cathy	Zulu	finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. find all unique departments from the employees table

* Syntax

```
SELECT DISTINCT  
department  
FROM employees;
```

department
IT
HR
Finance
Marketing

3. Retrieve the first and last names, ordered by salary from highest to lowest

* Syntax

```
SELECT first-name,  
last-name  
FROM employees  
ORDER BY salary DESC;
```

first-name	last-name
Eva	Naido
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees

* Syntax

```
SELECT id, first-name,  
       last-name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

Output

id	first-name	last-name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. find all employees who work in the IT department

* Syntax

```
SELECT id, first-name,  
       last-name, department  
FROM employees  
WHERE department = 'IT';
```

Output

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6 find employees in the finance department who earn more than 60 000

*Syntax

```
SELECT id, first-name, last-name,  
       department, salary  
FROM employees  
WHERE department = 'Finance' AND salary > 60 000;
```

Output

id	first-name	last-name	department	salary
3	Cathy	Zulu	finance	65000

7. Find all employees who work in the HR OR Marketing department

* Syntax

```
SELECT id, first-name,  
       last-name, department  
FROM employees  
WHERE department = 'HR' OR department = 'Marketing';
```

Output

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find all employees who are NOT in the IT department

* Syntax

```
SELECT id, first-name, last-name,  
       department  
FROM employees  
WHERE NOT department = 'IT';
```

Output

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

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9. find all employees who work in the IT, HR or Finance department

* Syntax

```
SELECT id, first-name, last-name,  
       department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance');
```

Output

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

10. find employees who:

1. Work in the IT department
2. Have a salary greater than 65 000 AND
3. Live in Johannesburg

* Syntax

```
SELECT id, first-name, last-name,  
       department, salary, city  
FROM employees  
WHERE department = 'IT' AND salary > 65 000  
       AND city = 'Johannesburg';
```

Output

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72006	Johannesburg